

BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

Name:

ID:

Due Date

Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `c:\Users\holly\4750 HW\hw3-ellasandwich`
Installed PrettyTables v3.0.11
Precompiling project...
17499.5 ms    PrettyTables
30820.3 ms    DataFrames
```

2566.3 ms Latexify → DataFramesExt
3 dependencies successfully precompiled in 75 seconds. 211 already precompiled.

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day⁻¹, and the decay rates of CBOD and NBOD are 0.35 and 0.25 day⁻¹, respectively. The river's saturated dissolved oxygen concentration is 10 mg/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance d km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m ³ /d	10,000 m ³ /d	15,000 m ³ /d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```

# constants
kc = 0.35
kn = 0.25
ka = 0.55
U = 6
Cs = 10

C_x = []
### Analytic Model, initial C0,B0,and N0
C0 = (100000*7.5+10000*5)/(110000)
B0 = (100000*5+10000*50)/(110000)
N0 = (100000*5+10000*35)/(110000)
# first 15m, before 2nd discharge
for i in 0:15
    a1 = exp(-ka*i/U)
    a2 = (kc/(ka-kc))*(exp(-kc*i/U)-exp(-ka*i/U))
    a3 = (kn/(ka-kn))*(exp(-kn*i/U)-exp(-ka*i/U))
    Cx = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
    push!(C_x, Cx)
end

# now, initial vars considering first discharge for 2nd
C_0 = (110000*C_x[16]+15000*5)/(125000)
B_0 = (110000*B0*exp(-kc*15/U)+15000*45)/(125000)
N_0 = (110000*N0*exp(-kn*15/U)+15000*35)/(125000)
Cx = C_0
# remaining 35 m (m15-m50) after 2nd discharge
for i in 16:50
    x = i-15
    a1 = exp(-ka*x/U)
    a2 = (kc/(ka-kc))*(exp(-kc*x/U)-exp(-ka*x/U))
    a3 = (kn/(ka-kn))*(exp(-kn*x/U)-exp(-ka*x/U))
    Cx = Cs*(1-a1) + C_0*a1 - B_0*a2 - N_0*a3
    push!(C_x, Cx)
end

min_SP = minimum(C_x)
println("Minimum DO, Streeter-Phelps (mg/L): $min_SP")

### Discretized model
C_x_disc = []
dx = 0.5
C0 = (100000*7.5+10000*5)/(110000)

```

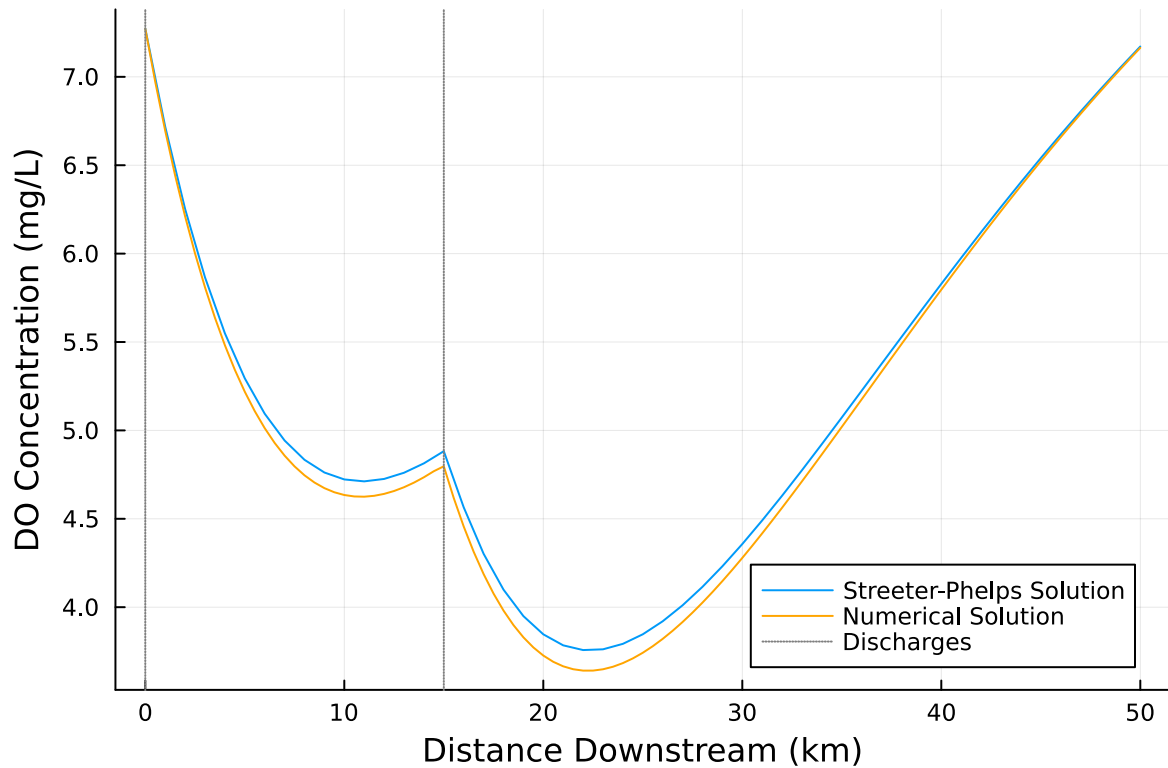
```

B0 = (100000*5+10000*50)/(110000)
N0 = (100000*5+10000*35)/(110000)
Cx = C0
for i in 0:dx:15-dx
    push!(C_x_disc, Cx)
    C_x1 = Cx + (dx/U) * (ka*(Cs-Cx) - kc*B0*exp(-kc*i/U)-kn*N0*exp(-kn*i/U))
    Cx = C_x1
end
C_0 = (110000*C_x_disc[end]+15000*5)/(125000)
B_0 = (110000*B0*exp(-kc*15/U)+15000*45)/(125000)
N_0 = (110000*N0*exp(-kn*15/U)+15000*35)/(125000)
Cx = C_0
for i in 15:dx:50
    x = i-15
    push!(C_x_disc, Cx)
    C_x1 = Cx + (dx/U) * (ka*(Cs-Cx) - kc*B_0*exp(-kc*x/U)-kn*N_0*exp(-kn*x/U))
    Cx = C_x1
end
min_Disc = minimum(C_x_disc)
println("Minimum DO, Numerical (mg/L): $min_Disc")
# graphing
x1 = [0:50]
p = plot(x1, C_x, label = "Streeter-Phelps Solution", xlabel= "Distance Downstream (km)", y1=0, y2=10)
x2 = [0:0.5:50]
plot!(x2, C_x_disc, color = "orange", label="Numerical Solution")
vline!([0,15], linestyle=:dot, label="Discharges", color = "grey")
display(p)

```

Minimum DO, Streeter-Phelps (mg/L): 3.757481773041718

Minimum DO, Numerical (mg/L): 3.6412522420697764



attempt to save state beyond implementation limit

Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

The numerical solution starts the same and pretty much ends the same as the streeter-phelps, but in between, the values are slightly below that of the streeter-phelps.

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the

4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

```
# Replication for solving problem 1.3, only stream 1
# constants
kc = 0.35
kn = 0.25
ka = 0.55
U = 6
Cs = 10
# percentage removal
p = 0.287
# Target DO: 4.2 mg/L

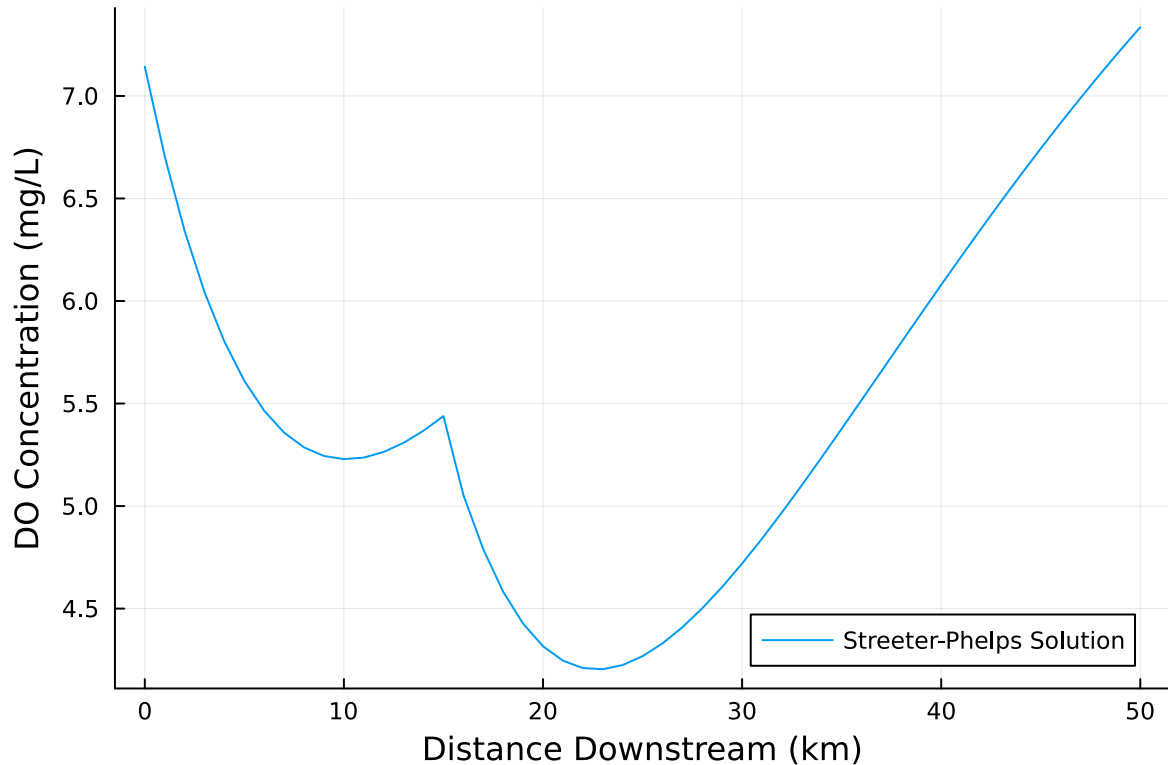
C_x = []
### Analytic Model, initial C0,B0,and N0
C0 = (100000*7.5+10000*5*(1-p))/(110000)
B0 = (100000*5+10000*50*(1-p))/(110000)
N0 = (100000*5+10000*35*(1-p))/(110000)
# first 15m, before 2nd discharge
for i in 0:15
    a1 = exp(-ka*i/U)
    a2 = (kc/(ka-kc))*(exp(-kc*i/U)-exp(-ka*i/U))
    a3 = (kn/(ka-kn))*(exp(-kn*i/U)-exp(-ka*i/U))
    Cx = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
    push!(C_x, Cx)
end
# getting value at x=15 before discharge
a1 = exp(-ka*15/U)
a2 = (kc/(ka-kc))*(exp(-kc*15/U)-exp(-ka*15/U))
a3 = (kn/(ka-kn))*(exp(-kn*15/U)-exp(-ka*15/U))
Cx_15 = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
# now, initial vars considering first discharge for 2nd
C_0 = (110000*C_x[16]+15000*5)/(125000)
B_0 = (110000*B0*exp(-kc*15/U)+15000*45)/(125000)
N_0 = (110000*N0*exp(-kn*15/U)+15000*35)/(125000)

# remaining 35 m (m15-m50) after 2nd discharge
for i in 16:50
    x = i-15
    a1 = exp(-ka*x/U)
    a2 = (kc/(ka-kc))*(exp(-kc*x/U)-exp(-ka*x/U))
    a3 = (kn/(ka-kn))*(exp(-kn*x/U)-exp(-ka*x/U))
```

```

Cx = Cs*(1-a1) + C_0*a1 - B_0*a2 - N_0*a3
push!(C_x, Cx)
end
min_SP = minimum(C_x)
println("Minimum DO, Streeter-Phelps (mg/L): $min_SP")
x1 = [0:50]
p = plot(x1, C_x, label = "Streeter-Phelps Solution", xlabel= "Distance Downstream (km)", y1=4.2, y2=7.5)
display(p)

```



Minimum DO, Streeter-Phelps (mg/L): 4.204477247747223

```

# Replication for solving problem 1.3, only stream 2
# constants
kc = 0.35
kn = 0.25
ka = 0.55
U = 6
Cs = 10

```

```

# percentage removal
p = 0.267
# Target DO: 4.2 mg/L

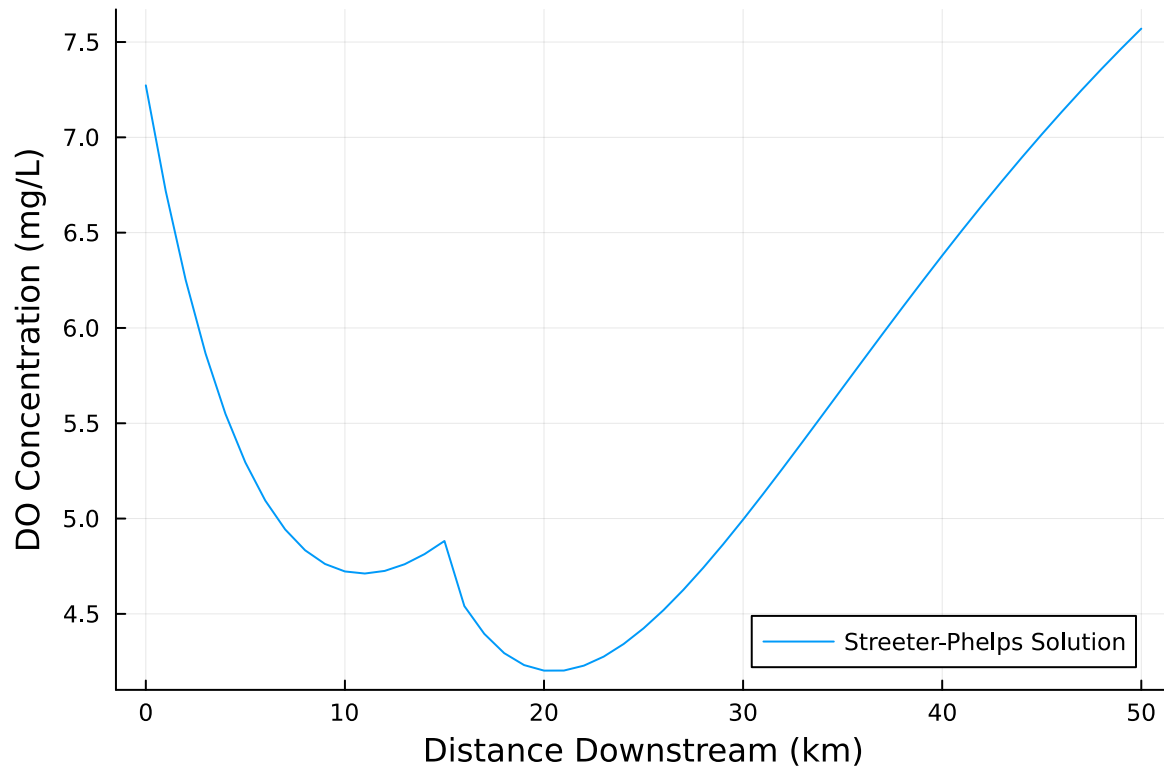
C_x = []
### Analytic Model, initial C0,B0,and N0
C0 = (100000*7.5+10000*5)/(110000)
B0 = (100000*5+10000*50)/(110000)
N0 = (100000*5+10000*35)/(110000)
# first 15m, before 2nd discharge
for i in 0:15
    a1 = exp(-ka*i/U)
    a2 = (kc/(ka-kc))*(exp(-kc*i/U)-exp(-ka*i/U))
    a3 = (kn/(ka-kn))*(exp(-kn*i/U)-exp(-ka*i/U))
    Cx = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
    push!(C_x, Cx)
end

# now, initial vars considering first discharge for 2nd
C_0 = (110000*C_x[16]+15000*5*(1-p))/(125000)
B_0 = (110000*B0*exp(-kc*15/U)+15000*45*(1-p))/(125000)
N_0 = (110000*N0*exp(-kn*15/U)+15000*35*(1-p))/(125000)

# remaining 35 m (m15-m50) after 2nd discharge
for i in 16:50
    x = i-15
    a1 = exp(-ka*x/U)
    a2 = (kc/(ka-kc))*(exp(-kc*x/U)-exp(-ka*x/U))
    a3 = (kn/(ka-kn))*(exp(-kn*x/U)-exp(-ka*x/U))
    Cx = Cs*(1-a1) + C_0*a1 - B_0*a2 - N_0*a3
    push!(C_x, Cx)
end
min_SP = minimum(C_x)
println("Minimum DO, Streeter-Phelps (mg/L): $min_SP")
x1 = [0:50]
p = plot(x1, C_x, label = "Streeter-Phelps Solution", xlabel= "Distance Downstream (km)", ylabel= "Dissolved Oxygen (mg/L)", display(p)

```

Minimum DO, Streeter-Phelps (mg/L): 4.2023796825997906



Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

You would have to remove less waste of stream 2 to meet the criteria, but only by a small amount (~27% vs ~29%). Therefore, more information is needed to determine which plan to choose. Cost of treatment for each would be beneficial, so we could determine which is cheaper and therefore more attractive a choice. If treating stream 1 is cheaper, you would most likely want to either treat only stream 1 or both streams equally.

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the

regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

```
# Replication for solving problem 1.5, Monte Carlo
# constants
kc = 0.35
kn = 0.25
ka = 0.55
U = 6
Cs = 10
# percentage removal
p = 0.287
# Target DO: 4.2 mg/L
min_array = []
inflow = LogNormal(2, 0.15)
len = 50000
in = rand(inflow, len)

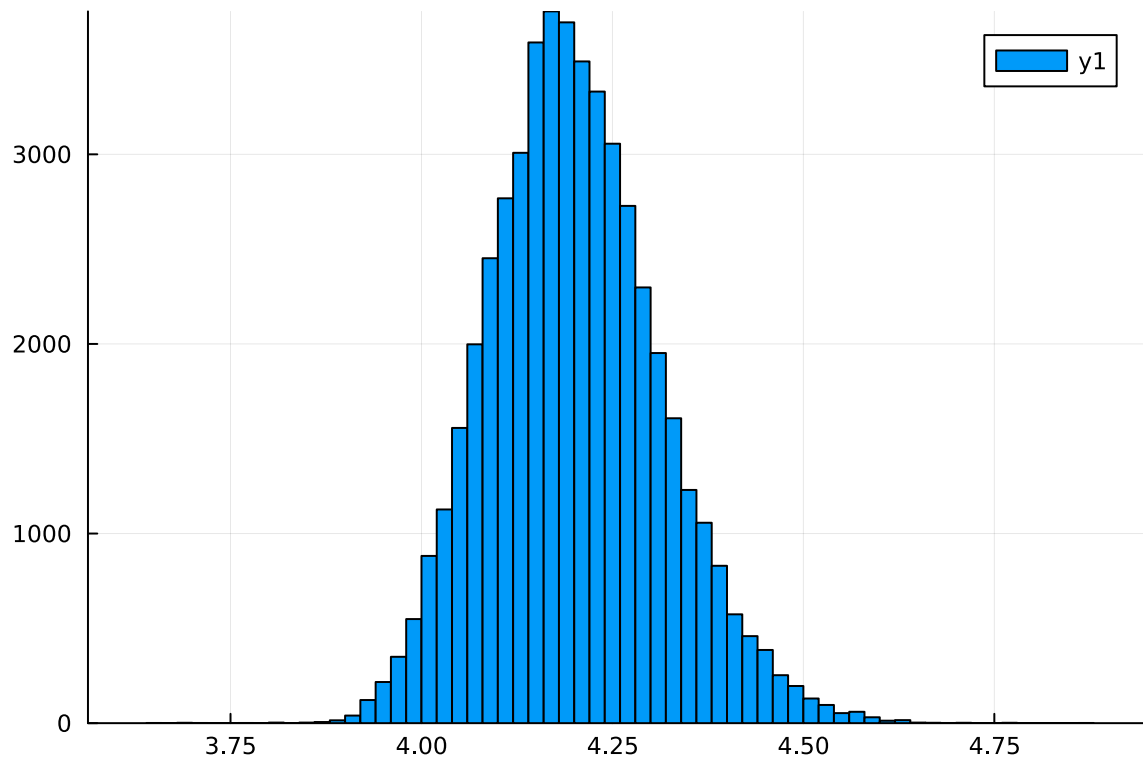
# Running Monte Carlo
for i in 1:length(in)
    C_x = []
    ### Analytic Model, initial C0,B0,and N0
    C0 = (100000*in[i]+10000*5*(1-p))/(110000)
    B0 = (100000*5+10000*50*(1-p))/(110000)
    N0 = (100000*5+10000*35*(1-p))/(110000)
    # first 15m, before 2nd discharge
    for i in 0:15
        a1 = exp(-ka*i/U)
        a2 = (kc/(ka-kc))*(exp(-kc*i/U)-exp(-ka*i/U))
        a3 = (kn/(ka-kn))*(exp(-kn*i/U)-exp(-ka*i/U))
        Cx = Cs*(1-a1) + C0*a1 - B0*a2 - N0*a3
        push!(C_x, Cx)
    end
    # now, initial vars considering first discharge for 2nd
    C_0 = (110000*C_x[16]+15000*5)/(125000)
    B_0 = (110000*B0*exp(-kc*15/U)+15000*45)/(125000)
    N_0 = (110000*N0*exp(-kn*15/U)+15000*35)/(125000)

    # remaining 35 m (m15-m50) after 2nd discharge
    for i in 16:50
        x = i-15
        a1 = exp(-ka*x/U)
        a2 = (kc/(ka-kc))*(exp(-kc*x/U)-exp(-ka*x/U))
```

```

        a3 = (kn/(ka-kn))*(exp(-kn*x/U)-exp(-ka*x/U))
        Cx = Cs*(1-a1) + C_0*a1 - B_0*a2 - N_0*a3
        push!(C_x, Cx)
    end
    min_SP = minimum(C_x)
    push!(min_array, min_SP)
end
# Plot histogram as sanity check
x = [0:len]
ave = mean(min_array)
pl = histogram(x, min_array, bins=100)
display(pl)
println("expected Value: $ave")
# Calculate Probability
no = 0
yes = 0
for i in 1:length(min_array)
    if min_array[i]<4.2
        no = no + 1
    else
        yes = yes + 1
    end
end
# probability
prob = no / (yes+no)
println("Probability of not meeting standard: $prob")
a = 0.05
ci = (quantile(min_array, a/2), quantile(min_array, 1 - a/2))
println("95% CI for random variable = $ci")

```



expected Value: 4.200154053236001

Probability of not meeting standard: 0.52284

95% CI for random variable = (3.9983156372516557, 4.439789587622377)

I determined my size for the Monte Carlo model by starting low and testing it, looking at the resulting histogram and the expected value. Starting at 1000, the expected value was fluctuating between values and the histogram was quite different each time. Continuing to test and raise this value, I reached a size of 50,000, which takes around 6 seconds to run, so it is not too large. The histogram remains pretty consistent, and the expected value only varies slightly in between runs. Since this sample size does not include any costs on the model, such as run-time, it is sufficiently large and not too large.

References

List any external references consulted, including classmates.